

Shubham Sirsat

Electronics & Telecommunication Engineering Student | UAV Systems & Embedded Design

Ahmednagar, Maharashtra, India | +91-9359649524 | shubhamsirsat011@gmail.com | [linkedin.com/in/shubham-sirsat](https://www.linkedin.com/in/shubham-sirsat)
github.com/sirsatshubham9359 | shubhamsirsat.in

Professional Summary

Final-year Electronics & Telecommunication Engineering student (SPPU) who engineered a Pixhawk-based quadcopter UAV achieving a **2.4:1** thrust-to-weight ratio and ± 1.5 m GPS position-hold accuracy, and built a closed-loop gimbal stabilization system that raised usable video quality from **45% to 92%**. Ranked in the **top 8%** of the Algo University Fellowship Program and won **1st Prize** at COGNOTSAV-2026, a state-level technical competition. Applies AI/data science techniques (RAG, LangChain) through internship and coursework.

Education

Bachelor of Engineering – Electronics & Telecommunication Engineering

Dr. Vithalrao Vikhe Patil College of Engineering, Ahmednagar – Savitribai Phule Pune University
SGPA: 9.59 (Sem 1) | 9.36 (Sem 2) | 8.77 (Sem 3) | 9.68 (Sem 4) | 8.86 (Sem 5) | 9.57 (Sem 6)

Graduated: 2027 (Expected)
Ahmednagar, India

Internship Experience

Drone Technology Intern

SSG Embedded Solutions, Nagpur

Feb 2026

- Covered UAV architecture, flight-controller fundamentals, and propulsion/power subsystem integration.

AI Data Science Intern

TechnoGrowth Software Solutions Pvt. Ltd.

31 Dec 2025 – 31 Jan 2026

- Applied AI/data science fundamentals – data analysis workflows and AI-driven solution design – to real-world problem sets.

Embedded Systems Design Intern

SkillDzire – AICTE Approved Internship Programme

Aug 2025 (3 Weeks)

- Built microcontroller-based hardware, designed circuits, and implemented sensor interfacing.

Projects

Quadcopter Drone with 2-Axis Gimbal Camera System

2025 – 2026

Dr. Vithalrao Vikhe Patil College of Engineering (SPPU) – Team Project

- Engineered and assembled a Pixhawk 2.4.8-powered quadcopter UAV with brushless DC motors, ESCs, and a 3S LiPo battery, achieving a **2.4:1** thrust-to-weight ratio.
- Designed a closed-loop 2-axis gimbal stabilization system, raising usable video frame quality from **45% to 92%** and holding GPS position within ± 1.5 m.
- Calibrated the full flight-control stack via Mission Planner, monitoring live **telemetry** data throughout tuning; won **1st Prize at COGNOTSAV-2026**, a state-level technical paper & project competition.

Campus Authentication Concept – Contactless Login System

2026

Personal Project

- Designed a concept and system architecture for replacing password logins with time-limited QR codes and mobile biometric verification; used AI-assisted development tools to build a working prototype.

GPS-Based Autonomous Delivery Drone (Final Year Project – In Progress)

2026 – 2027

Savitribai Phule Pune University

- Developing an autonomous GPS-guided delivery drone on existing Pixhawk hardware, paired with a real-time **telemetry** tracking dashboard for live flight and delivery status.

Technical Skills

Languages: Python, C, C++, Core Java

Developer Tools: Git, GitHub, MATLAB, Mission Planner, Microsoft Office (Word, Excel, PowerPoint)

Technologies / Frameworks: Pixhawk Flight Controller, PX4 Firmware, UAV/Drone Systems, Embedded Systems, GPS & Sensor Integration, PID Control, CAN Bus, Telemetry Protocols, Simulink, LangChain, Retrieval-Augmented Generation (RAG)

Core Strengths: Team Leadership, Cross-Functional Collaboration, Technical Documentation, Problem Solving

Certifications

- 5-Day AI Agents Intensive Course** – Google & Kaggle | **Getting Started with Cybersecurity** – IBM SkillsBuild | **Getting Started with AI** – IBM SkillsBuild
- Journey to Cloud** – IBM SkillsBuild | **RAG with LangChain** – IBM SkillsBuild | **MATLAB Onramp** – MathWorks
- Python for Data Science, AI & Development** – IBM/Coursera | **Data Analytics Simulation** – Deloitte | **GenAI Analytics Simulation** – Tata

Awards & Leadership

- 1st Prize**, COGNOTSAV-2026 State-Level Technical Paper & Project Competition | **Top 8%**, Algo University Fellowship Program
- Vice-President**, EESA | **Departmental Coordinator**, National Service Scheme (NSS) – Dr. Vithalrao Vikhe Patil College of Engineering
- Represented **District-Level Kabaddi Team** | A Grade, Elementary & Intermediate Examinations